Module - I

Cell Theory
Cell Structure & Function

R. K. Gothwal, PhD
Department of Bioengineering and Biotechnology
BIT Mesra, Ranchi
Jaipur-Campus

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Cell

- In 1665, the first cell was discovered and named by **Robert Hooke** while observing dead cork samples with a crude lens. He was an English scientist and architect who, using a microscope, was the first to visualize a microorganism.

Robert Hooke
British Scientist

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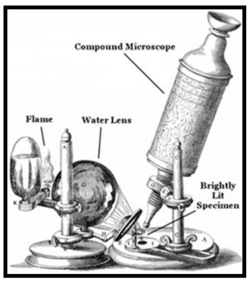

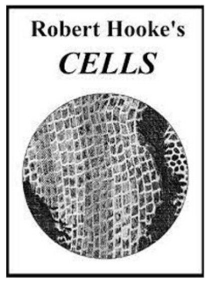

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Cell

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- **Antonie van Leeuwenhoek** in 1674 first saw and described **<u>a live cell</u>** in algae Spirogyra. He was a Dutch businessman and scientist in the Golden Age of Dutch science and technology. A largely self-taught man in science, he is commonly known as "**the Father of Microbiology**", and one of the first microscopists and microbiologists.

Robert Hooke
British Scientist

Antonie van Leeuwenhoek
Dutch Scientist

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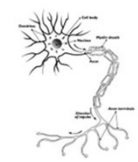
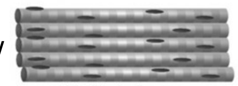
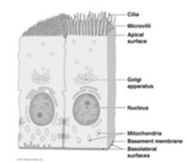
Cell

- All living organisms are made up of cells.
- The cell is the smallest unit of matter that is alive and is called as structural and functional unit of life.
- Each cell has amazing world in itself. It takes in nutrients, convert these nutrients into energy, carry out specialized functions, and they grow with time and self-propagate by reproduction for survival.
- Each cell differentiates into specific cell type for playing a specific role in the life of an organism. Means, specific organs for specific function e.g. Cardiac, Hepatic, Nerve, Myeloma etc.







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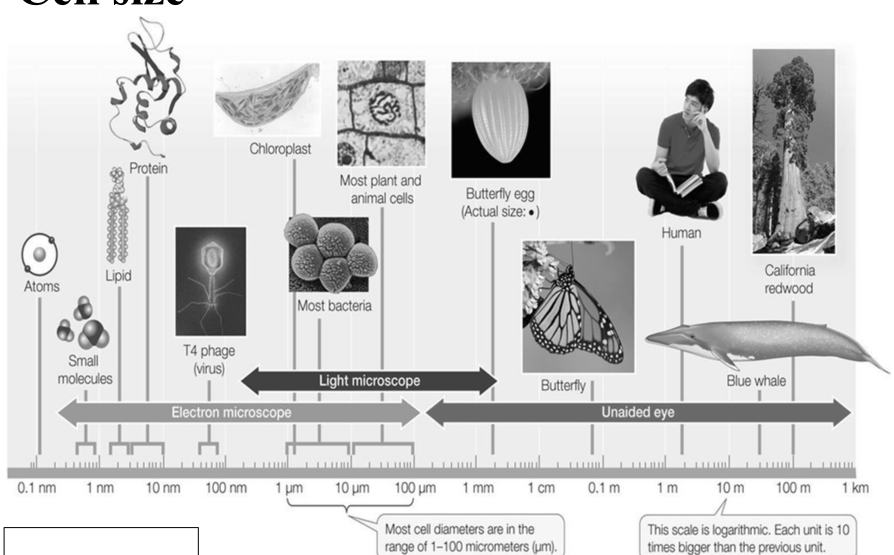
Cell size



Purves *et al*. (1998). Life the Science of Biology, Sinauer Associates, Inc.

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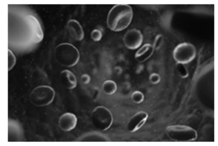
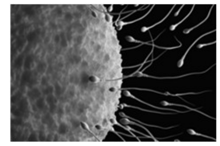
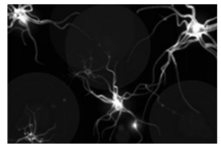
Cell Types

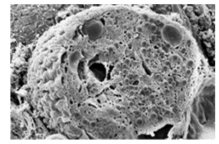
In Humans - More than 200 cells types

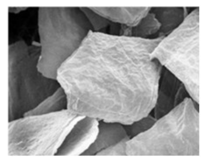
- Blood Cells (RBC,WBC)
- Stem Cells
- Nerve Cell (Neuron)
- Muscle Cell (Myocyte)
- Skin Cells (Epithelial)
- Bone cell (Osteocytes)
- Fat Cell (Adipocytes)
- Cartilage cells (chondrocytes)
- Sex Cells (Gametes)

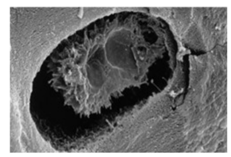

Blood cells


Sex cells


Nerve cells


Pancreatic cells


Skin cells


Bone cells

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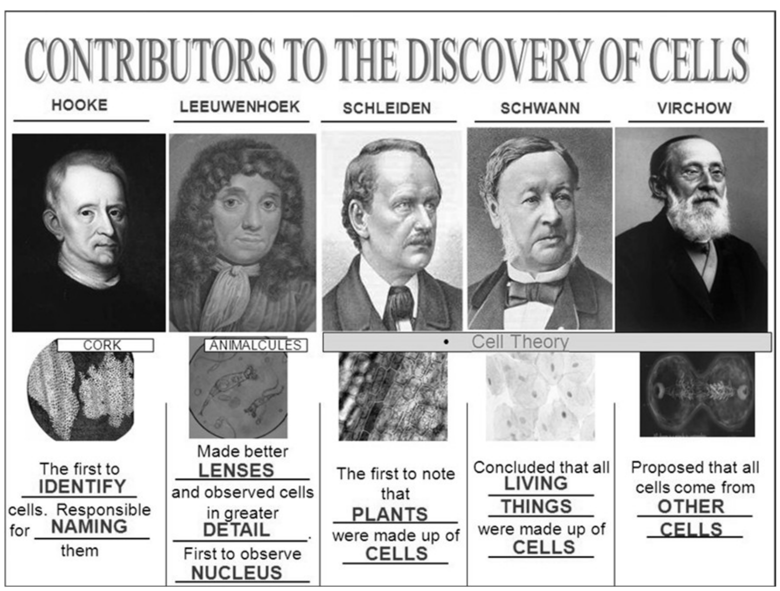

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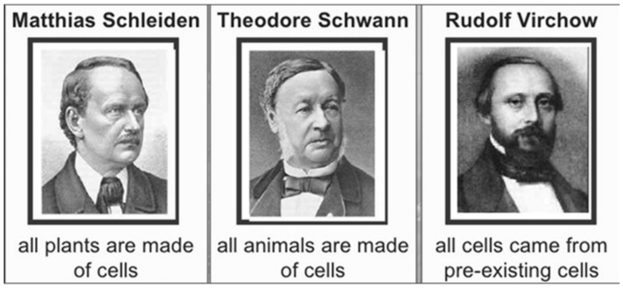
Cell Theory

Three components:

- ✓ All organisms are composed of one or more cells (Schleiden and Schwann, 1938-39).
- ✓ The Cell is the basic unit of life (Schleiden and Schwann, 1938-39).
- ✓ All cells are produced by division of pre-existing cells (Virchow 1858).



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Modern Cell Theory

- ❖ All known living things are made up of cells.
- ❖ The cell is a structural & functional unit of all living things.
- ❖ All cells come from pre-existing cells by division and spontaneous generation does not occur.
- ❖ Cells contain hereditary information which is passed from one cell to the daughter cell during the cell division.
- ❖ All cells are basically the same in chemical composition.
- ❖ All energy flow (metabolism) of life occurs within the cells.

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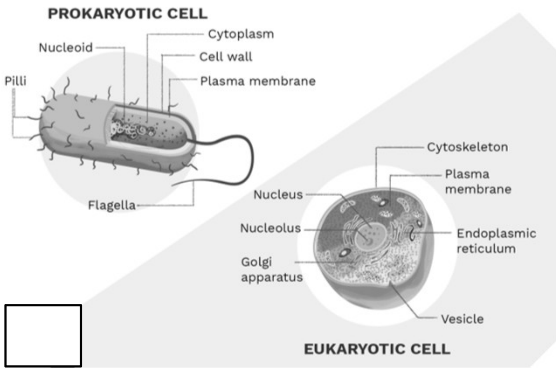
Prokaryotic and Eukaryotic Cells

- **Prokaryotic cells** are very diverse in their metabolic capabilities. They can live off a greater diversity of energy sources than any other living creatures, and they inhabit greater environmental extremes.
- **Animals, plants, fungi and protists** have cells that are larger (10 T) and structurally more complex than those of prokaryotes.



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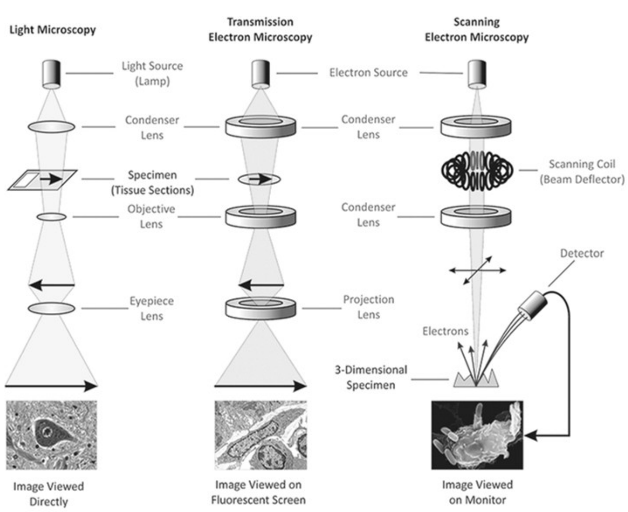
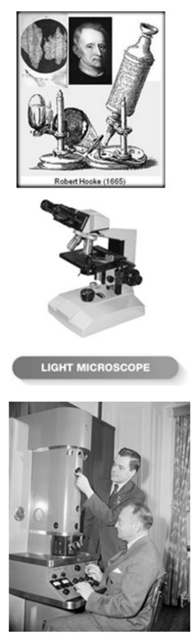
Cell Structure - Microscopy





Ernst Ruska and Max Knoll - 1931

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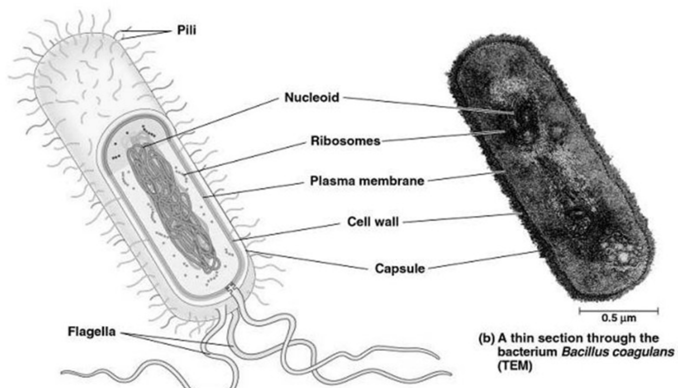
Prokaryotic Cells

A typical bacterial cell is composed of Cell wall, Plasma membrane, Nucleoid, Ribosomes, Pilli, Flagella, storage granules and a Capsule. Not all structures are present in all prokaryotic cells; it may vary from one species to other.



(a) A typical rod-shaped bacterium

(b) A thin section through the bacterium *Bacillus coagulans* (TEM)

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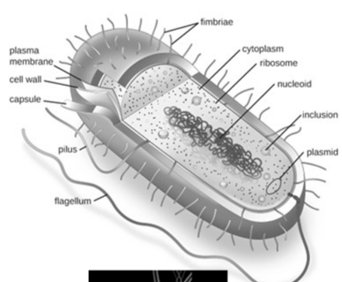
Prokaryotic Cells

- **Capsule (or slime layer)**
  - A thick polysaccharide layer outside the cell wall.
  - Not present in all bacteria. Found only in some gram +ve bacteria, if a capsule is present, then flagella are not.
  - Used for sticking cells together, as a food reserve, as protection against desiccation and chemicals and as protection against phagocytosis.
- **Flagella (singular flagellum)**
  - A thin and long filamentous extension or whip-like structure that aids in cellular locomotion.
  - It is extended from the cell wall and used for the motility of bacteria.
  - Bacterial flagellum is composed of three parts- filament, hook and basal body.
- **Pilli and Fimbriae**
  - Hair-like structures on the surface of the cell that attach to other bacterial cells. Shorter pili called fimbriae help bacteria attach to surfaces.



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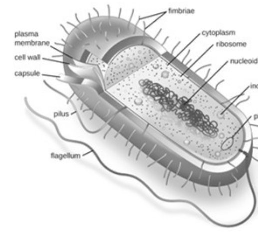
Prokaryotic Cells

- **Cell Wall**

Made up of murein (not cellulose), which is a glycoprotein (also called peptidoglycan). There are two kinds of cell wall, which can be distinguished by Gram staining -
- **Gram +ve** bacteria have a thick cell wall, stain purple, may have spores and are sensitive to penicillin and lysozymes (an antibacterial enzyme found in tears and saliva)
- **Gram –ve** bacteria have a thin cell wall with an outer lipid layer, have no spores and stain pink, these are thought to be more highly evolved.



• **Plasma/Cell/Selectively permiable membrane**
Boundary (“wall”) between the cell and the environment. Allows nutrients/regulates movement in and out of the cell. Made up of phospholipids and proteins.

• **Cytoplasm**
Cytoplasm is a jelly-like substance that is sometimes described as "the cell-matrix". Contains all the enzymes needed for all the metabolic reactions.

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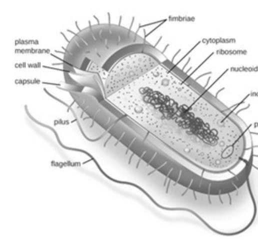
Prokaryotic Cells

• **Nucleoid**

The region of the cytoplasm that contain DNA. It is not surrounded by nuclear membrane. Sometimes also referred as nuclear body or nuclear zone.



• **Plasmid**

Small circular DNA, used to exchange DNA between bacterial cells. Used in genetic engineering, they often contain genes giving resistance against antibiotics.

• **Ribosomes**

The smaller 70S type. Two units 50S and 30S. They are the site of protein synthesis. Several ribosomes may attach to a single mRNA and form a chain called polyribosomes or polysome. The ribosomes of a polysome translate the mRNA into proteins.

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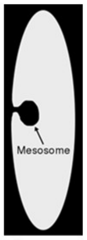
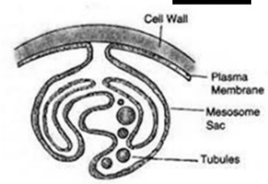
Prokaryotic Cells



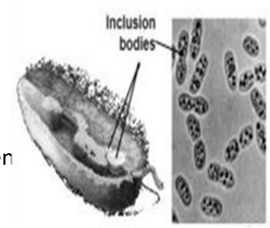
- **Mesosomes**

- Special intracellular structure which is formed by the extensions of plasma membrane into the cell. These extensions are in the form of vesicles, tubules and lamellae. They help in cell wall formation, DNA replication and distribution to daughter cells.
- They also help in respiration, secretion processes, to increase the surface area of the plasma membrane and enzymatic content. In some prokaryotes like cyanobacteria, there are other membranous extensions into the cytoplasm called chromatophores which contain pigments.



- **Storage Granules (or Inclusion bodies)**

- Reserve materials in prokaryotic cells are stored in the cytoplasm in the form of inclusion bodies. These are not bound by any membrane system and lie free in the cytoplasm, e.g. in the form of glycogen, lipids, polyphosphate, or in some cases sulfur or nitrogen.
- Gas vacuoles are found in blue green and purple and green photosynthetic bacteria.



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THANK YOU

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